GLOBALMART SALES REPORT

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Business understanding

Business Context

GlobalMart is a multinational retail company operating across multiple regions and product categories, including Furniture, Office Supplies, and Technology. The company seeks to improve sales performance and profitability through data-driven decision-making.

Business Problem

1. Which regions and states drive the most revenue?
2. Which product categories and sub-categories contribute most to profit?
3. The impact of discounting on profitability?
4. Operational efficiency, including shipping performance?

Business Objectives

The primary objectives of this analysis are:

1. Evaluate overall sales and profit performance.
2. Identify high-performing and underperforming regions.
3. Detect profit leaks at the sub-category level.
4. Assess the relationship between discounts and profit margins.
5. Provide actionable recommendations to improve profitability and growth.

Data understanding

Dataset includes 1010 rows & variables as below.

- **Order ID**: Unique identifier for each order -Text
- **Order Date**: Date the order was placed -Date
- **Ship Date**: Date the order was shipped -Date
- **Region**: Geographical region (e.g., East, West, Central, South) -Text
- **State**: State within the region -Text
- **Category**: Product category (e.g., Furniture, Office Supplies, Technology) -Text
- **Sub-Category**: Sub-category of the product (e.g., Chairs, Binders, Phones) -Text
- **Product Name**: Name of the product -Text
- **Sales**: Revenue generated from the sale -Number/Currency
- **Quantity**: Number of units sold -Number
- **Discount**: Discount applied (in percentage) -Percentage / decimal number
- **Profit**: Profit earned from the sale -Number/Currency

Data preparation

Data preparation involved type validation, removal of incomplete transactions (missing Order Date, Region, and Sales), correction of inconsistent text formatting, exclusion of anomalous negative revenue entries, and creation of derived variables including Shipping Days and Discount Bands.

A dedicated Date table was constructed to enable time-based analysis.

Modelling

A star schema data model was implemented to optimize analytical performance and ensure scalable design.

I created the tables below:

- **DimDate**: To enable consistent time slicing and time intelligence calculations.
- **DimProduct**: To normalize product attributes to support hierarchical drilldown and efficient filtering.
- **DimGeo**: To normalize geography to support Region to State analysis and mapping.
- **DimDiscountBand** To support interactive discount impact analysis without distorting the star schema.

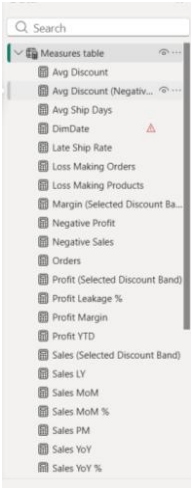
All relationships were configured as Many-to-One with single-direction filtering (Dimension - Fact), ensuring consistent filter propagation and preventing ambiguity in calculations.

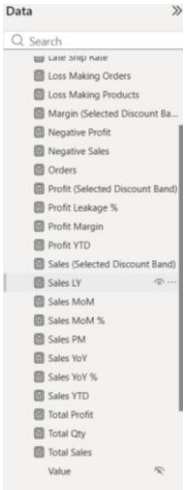
The DimDate table was marked as the official date table to enable time intelligence functions such as Year-to-Date (YTD), Month-over-Month (MoM), and Year-over-Year (YoY) calculations.

Evaluation

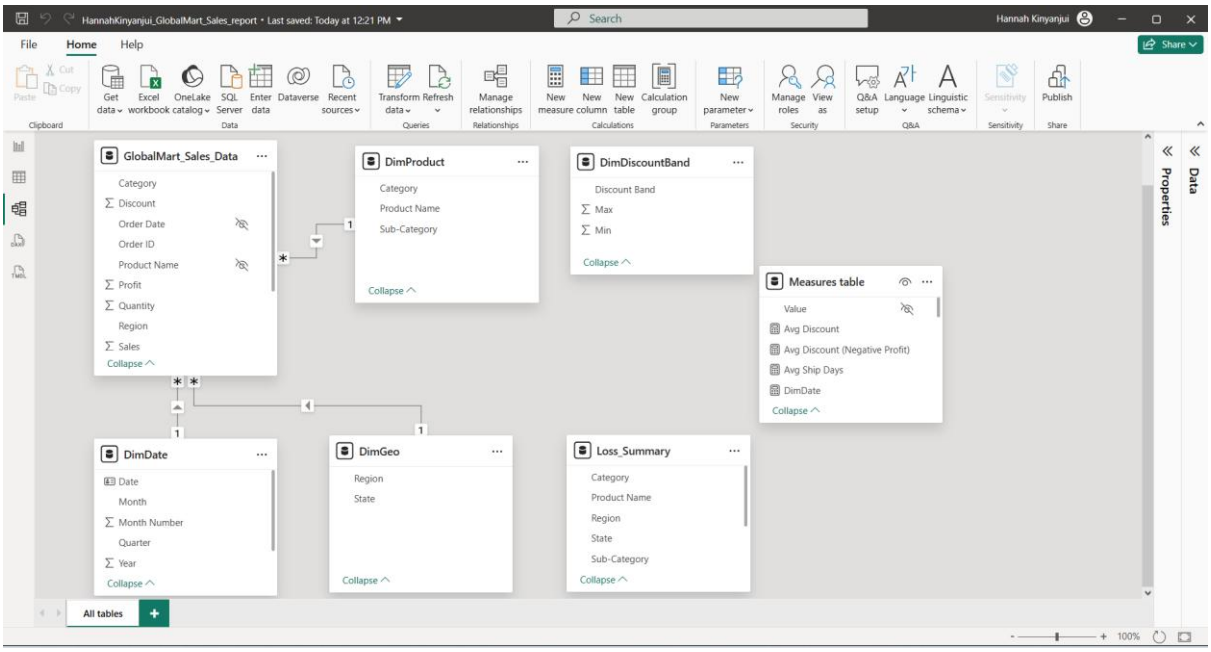
The evaluation phase focused on validating whether the developed Power BI solution met the business objectives defined at the start of the project. The primary objective was to analyze GlobalMart's sales performance to identify growth opportunities, profitability drivers, and operational inefficiencies.

First, KPI calculations were validated against raw totals to ensure numerical accuracy. Measures such as Total Sales, Total Profit, Profit Margin, YTD Sales, YoY Growth, and MoM Growth were tested under different slicer conditions (Year, Category, Region, Discount Band) to confirm that filter context behaved correctly. Time intelligence measures were verified using the dedicated date table to ensure correct period comparisons.

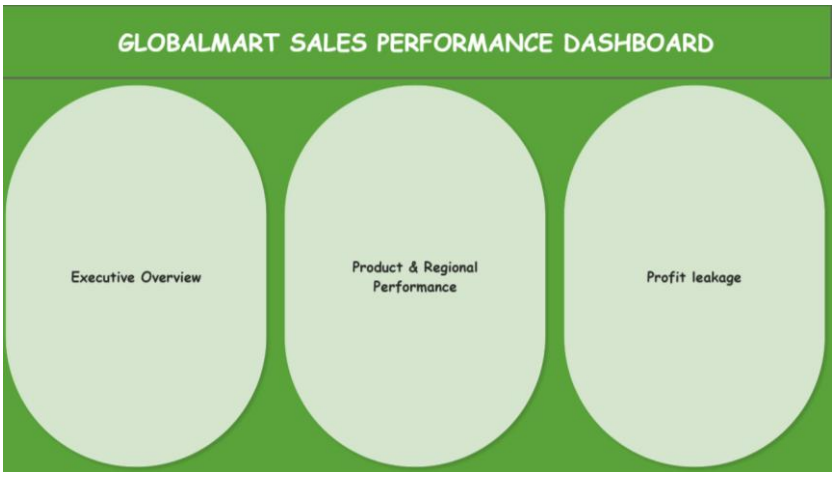




Second, the data model was reviewed to ensure proper star schema structure. Relationships were validated to prevent circular dependencies and ambiguous filter paths. Cross-filter direction was kept single-directional to optimize performance and maintain analytical integrity.



Third, visual outputs were examined to confirm alignment with business questions:

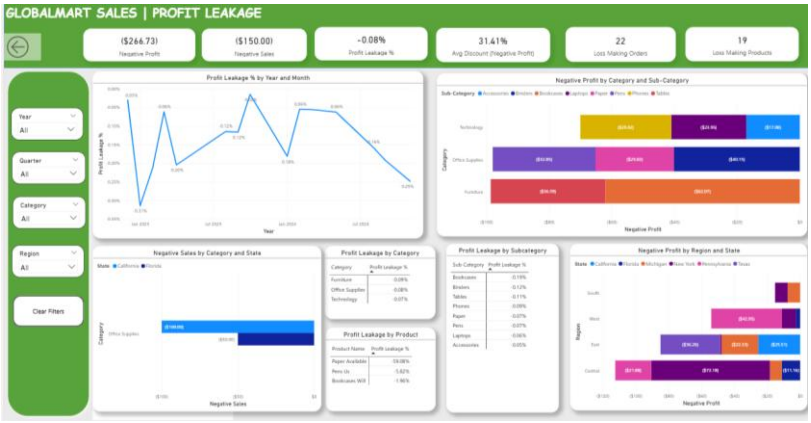


- Executive Summary provided a high-level financial overview.

- Product Performance highlighted revenue drivers and margin variability.

- Regional Analysis identified geographic performance differences.

- Profit Leakage isolated loss-making products and discount impact.



Outliers and negative profit records were intentionally retained to allow diagnostic analysis rather than artificially inflating performance metrics.

Findings

Overall Financial Performance

GlobalMart generated $1.08M in total sales and $326.21K in profit, delivering a strong 30.23% profit margin. This indicates a healthy business overall, but performance is not stable over years. In 2023, sales reached $834.14K with $156.40K profit, while 2024 shows a sharp decline in growth (YoY -72.74%), despite maintaining a high margin (69%). This suggests the business shifted from volume-driven growth to lower-volume, high-margin transactions.

Operationally, efficiency is a concern. The late shipment rate is consistently high (41%), with average shipping around 4.5 days, indicating delays across regions. Additionally, profit leakage exists (up to -0.10%), driven by discount-heavy transactions (30–31%).

Overall, the business is profitable but losing efficiency through operations and discounting.

Executive Overview

The company demonstrates strong profitability at aggregate level, with consistent margins around 30% overall, and even higher 69% in 2024. Sales peaked in 2023, supported by strong category performance, but declined significantly in 2024, indicating inconsistent revenue generation. This suggests reliance on specific high-performing periods rather than sustained growth.

Monthly trends show fluctuations, with spikes and drops in sales, reinforcing that demand is not evenly distributed. While profit margins remain strong, the presence of negative profit transactions and declining sales trends indicates that profitability is not uniformly distributed across the business, requiring tighter control on pricing and sales strategy.

Product & Category Performance

Office Supplies is the leading category, generating the highest sales ($0.6M) and strong profit contribution, followed by Furniture and Technology. High-performing sub-categories such as Binders (71%), Chairs (70%), and Pens (69%) drive profitability, making them critical to business success.

However, performance is uneven. Sub-categories like Paper (0.6% margin) and Tables (22%) significantly underperform despite contributing to sales. The product table further shows negative profit margins for several items (e.g., Tables -124%, Bookcases -77%), indicating that some products are being sold at a loss, largely due to discounts. This highlights inefficient product-level pricing strategy.

Regional Performance

The West region is the dominant contributor (67% of total sales), making it the primary revenue driver. Other regions (East, Central, South) contribute significantly less, indicating high geographic concentration risk. States such as California, New York, and Texas consistently appear as top contributors in both sales and profit.

However, inefficiencies are visible across regions. The Central region shows the highest negative profit values, while all regions experience high late shipment rates (40–50%), including top-performing states. This suggests that strong sales performance is being offset by operational inefficiencies, particularly logistics delays, which may impact profitability and customer experience.

Profit Leakage Diagnostics

Profit leakage is relatively small overall (-0.08% to -0.10%), but highly concentrated. Furniture (especially Tables) is the largest contributor to losses, followed by Office Supplies and Technology sub-categories. There are 22 loss-making orders and 19 loss-making products, confirming that leakage is recurring and not incidental.

The primary driver is clearly high discounting, with average discounts on negative-profit transactions reaching 30–31%. Leakage is also regionally concentrated, particularly in the Central region, which shows the highest negative profit values. This indicates a combination of pricing inefficiencies and regional operational challenges.

Recommendations

1. Introduce discount caps (20–25%) to prevent excessive margin erosion.
2. Implement margin-based pricing controls, blocking sales below profitability thresholds.
3. Focus on scaling high-margin sub-categories (Binders, Chairs, Pens).
4. Reprice or discontinue low-margin products (Tables, Paper).
5. Reduce dependency on the West region by strengthening underperforming regions.
6. Improve logistics to reduce late shipment rate (41%), especially in high-delay states.
7. Bundle low-margin products with high-margin ones to improve overall transaction profitability.
8. Create automated alerts for loss-making orders/products within Power BI.
9. Create an automatic data refresh setting to keep the dashboard up to date